NTDH: Complex Reasoning for Comprehensive Affective Analysis

Tianlei Zhu[a], Zhiwei Liu[b,*], Yuyan Wang[b], Xiao-Yang Liu[a] and Sophia Ananiadou[b]

[a]Columbia University, New York, NY, USA
[b]Department of Computer Science, The University of Manchester, Manchester, United Kingdom

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ABSTRACT

Comprehensive affective analysis is challenging for two reasons: it spans heterogeneous prediction tasks with continuous, ordinal, and multi-label outputs, and affective meaning is context-dependent, requiring conflicting cues to be reconciled rather than mapped directly to labels. Existing methods learn this mapping directly and do not model the reconciliation explicitly. We recast the task as a *complex-reasoning* problem, which yields one output interface across heterogeneous label spaces and a trajectory over which a verifiable reward can be optimised; to our knowledge, this is the first such treatment covering both sentiment and emotion. The obstacle is on the data side: affective reasoning traces must be synthesised, and generic synthesis is misaligned with the targets, tolerances, and phenomena of affect, and discards or leaks its failure cases. We propose **NTDH**, which addresses these four failures. *Naturalisation* sets the training answer to the gold label, so it is correct by construction. A *Tolerance*-aware gate checks each answer against the task's own scoring margin. *Domain*-aware strategies refine the reasoning using ideas from affective science. Directional *Hints* report only the type and direction of an error, without exposing the target. We train Qwen3-8B with SFT and then GRPO under the same tolerance used for verification (up to a more permissive construction gate on the multi-label subtask), and a component ablation quantifies the data-quality effect of each part. Using 16,302 training records, about 14× fewer than comparable instruction-tuned systems, the final policy improves over its SFT checkpoint on five of six official-test metrics and achieves the strongest EI-reg result among the compared systems, at a Pearson correlation of 0.862.

1. Introduction

Affective analysis aims to identify the sentiment, emotions, and affective intensities expressed in text, providing fundamental signals for applications such as opinion and stance analysis, emotion-aware retrieval, conversational agents, mental-health monitoring, and social-media analysis [1]. Despite substantial progress, comprehensive affective analysis remains challenging for two fundamental reasons. First, it encompasses heterogeneous prediction tasks. Sentiment and emotion capture complementary dimensions of affect, while outputs may be represented as continuous intensities, ordinal categories, or multi-label emotion sets [2]. Consequently, heterogeneous labels require unified representations while preserving their task-specific semantics and evaluation criteria. Second, affective meaning is inherently context-dependent. Reliable prediction requires integrating multiple contextual cues, including polarity shifts (e.g., negation and intensification) [3, 4], figurative language (e.g., irony, sarcasm, and idioms) [5, 6], and interactions among multiple emotions [7–9], rather than directly mapping text to labels. We investigate these challenges across the four complementary SemEval-2018 subtasks: emotion-intensity regression (EI-reg), multi-label emotion classification (E-c), valence regression (V-reg), and ordinal valence classification (V-oc) [10].

Research on affective analysis has evolved from feature-based methods [11–13] through pre-trained language models to instruction-tuned large language models (LLMs), of which EmoLLM [14] is the closest to our setting; §2.1 reviews this line in detail. Nevertheless, existing methods primarily learn a direct mapping from input text to output labels, without explicitly modelling the reasoning process required to reconcile contextual and potentially conflicting affective evidence.

Modelling this process explicitly is valuable less for immediate accuracy gains than for three capabilities that a direct mapping cannot provide. First, it supplies a single output interface for heterogeneous label spaces: continuous intensities, ordinal categories, and multi-label emotion sets can all be expressed as a structured reasoning trace terminating in a natural-language conclusion, so that affective cue analysis is shared across subtasks and only the final commitment remains task-specific. This makes a single policy, rather than one specialised model per output space, feasible for comprehensive affective analysis. Second, affective labels are verifiable under task-specific criteria,

*Corresponding author. E-mail address: zhiwei.liu@manchester.ac.uk

namely numerical tolerance for regression, exact match for ordinal classes, and set agreement for multi-label outputs. Verifiability is a precondition for reward-based optimisation, but a reward requires a space over which to optimise; an explicit reasoning trace provides that space, whereas a single scalar prediction offers reinforcement learning nothing to reshape. Third, an explicit trace renders the judgement inspectable, which is important both for a task whose annotations are themselves contested and for deployment settings such as mental-health monitoring, where an unexplained score is not actionable.

Recent reasoning frameworks combining supervised reasoning traces with reinforcement learning have demonstrated remarkable success in domains such as mathematics [15, 16] and clinical question answering [17], where rationales can be synthesised by a teacher model and filtered by a judge [18, 19]. However, directly transferring these frameworks to affective analysis remains challenging. Generated reasoning traces may terminate in incorrect or inconsistently formatted conclusions, generic verification mechanisms cannot faithfully evaluate heterogeneous regression and classification tasks, domain-independent refinement offers limited support for affective phenomena such as valence shifting, irony, and emotion interactions, and unsuccessful trajectories are commonly discarded or repaired by revealing the gold answer. These failures define four requirements for constructing reliable affective reasoning data: (1) task-consistent target representations, (2) evaluation-aware verification, (3) domain-informed reasoning, and (4) informative guidance for resolving difficult cases without compromising supervision. Existing reasoning frameworks fail to satisfy these four requirements simultaneously.

To address these limitations, we propose **NTDH**, a *quality-aware reasoning-data synthesis framework* for cross-task affective analysis. Rather than treating initially generated reasoning traces as ready-to-use supervision, NTDH controls four dimensions of reasoning-data quality through four complementary components, one per requirement. **Naturalisation (N)** converts heterogeneous task labels into scale-aware natural-language targets while preserving their task-specific semantics. **Tolerance-aware Verification (T)** evaluates generated conclusions using task-specific acceptance criteria that account for numerical tolerances, categorical decisions, and multi-label agreement. **Domain-aware Refinement (D)** revises unsuccessful reasoning trajectories using affect-specific knowledge, including contextual valence shifts, figurative language, and interactions among emotions (§3). **Directional Hints (H)** provide target-derived corrective feedback without directly disclosing the exact reference answer, enabling difficult trajectories to be refined while reducing direct answer leakage.

Using the resulting quality-controlled corpus, we first perform supervised fine-tuning (SFT) to initialise Qwen3-8B [20] with cross-task affective reasoning patterns. We subsequently apply Group Relative Policy Optimization (GRPO) [17, 21] with task-specific verifiable rewards, defined by the same tolerance the verifier applies, to further optimise its reasoning and prediction policy. Together, these components provide a unified approach to constructing and exploiting reasoning supervision across heterogeneous affective tasks.

Our contributions are:

1. **A unified reasoning formulation for affective analysis.** We formulate sentiment, emotion, intensity regression, ordinal prediction and multi-label classification as structured generation with an explicit reasoning path, rather than as independent text-to-label mappings, obtaining one output interface and one verifiable reward across four heterogeneous subtasks.
2. **NTDH, a quality-aware reasoning pipeline.** Four data-side fixes, each grounded in the affect literature: *Naturalisation*, *Tolerance*-aware verification, *Domain*-aware refinement, and directional *Hints*. Only 18.4% of the initial model-generated conclusions satisfy the gold tolerance; Naturalisation instead supplies a gold-consistent <answer> for every retained SFT trace. Of the half-A pool, 5,388 verified traces form the SFT set, while the remaining 2,762 hard cases are routed to RL with half-B. This routing uses all 16,302 instances (5,388 SFT + 10,914 RL). A component ablation (§5.3) checks each part: N drives target correctness, T removes 48–63% of the out-of-tolerance accepts, H eliminates label leakage from the fallback, and D helps the fine-grained E-c subtask.
3. **Evaluation across four affective tasks.** Using 16,302 total SFT and RL records—about 14× fewer than EmoLLM's 234K instruction records—RL-final outperforms the SFT initialisation checkpoint on five of six metrics on the official 9,201-instance test set and achieves the strongest EI-reg result among the compared systems, while remaining competitive across all four subtasks. Component ablations and qualitative error analysis identify where performance changes and where rare-emotion recall remains limiting.

2. Related Work

2.1. Affective Analysis of Text

Recent affective-analysis research has shifted from task-specific encoders to generative language models [1]. Transformer encoders such as BERT [22] and RoBERTa [23], together with affect- and social-media-specialised variants including BERTweet [24] and SentiBERT [25], remain strong supervised baselines, but are typically fine-tuned with a separate prediction head for each label space. This task-specific paradigm also characterises the original SemEval-2018 Task 1 systems [10]: SeerNet combines multiple lexical regressors [26], whereas NTUA-SLP uses deep attentive recurrent networks [27].

Work since 2023 has increasingly examined whether LLMs can replace these separate pipelines with prompting or instruction tuning. A broad evaluation over 13 sentiment-analysis tasks and 26 datasets finds that LLMs are effective on simpler settings but still lag specialised models on several complex or structured tasks [28]. Recent systems therefore target particular sources of difficulty: THOR elicits reasoning for implicit sentiment [29]; InstructERC combines instruction tuning and retrieval for conversational emotion [30]; Feng et al. compare in-context learning and task-specific fine-tuning for conversational affect [31]; DECC decomposes emotion-cause reasoning [32]; and TEII develops iterative LLM workflows for cross-lingual emotion detection [33].

Other recent work broadens either task coverage or interpretability. EmoLLM unifies multiple affective tasks through large-scale instruction tuning [14]; EmotionQueen evaluates event, implicit-emotion, intention and empathetic-response capabilities [34]; and MASIVE moves beyond a small closed inventory toward open-ended affective-state generation [35]. EmoRationale adds retrieved evidence to multilingual emotion and intensity predictions [36], while cognitive-science-based analyses argue for grounding model design in theories of emotion and communication [37]. Nevertheless, these approaches mainly address a single affective setting, evaluate general-purpose LLMs, or generate answers without constructing verifier-controlled reasoning trajectories across sentiment, emotion and intensity. NTDH addresses this gap while retaining EmoLLM's unified, instruction-following formulation.

2.2. Complex Reasoning with LLMs

A second line of work makes LLMs reason step by step. Chain-of-thought (CoT) prompting elicits intermediate reasoning and improves multi-step tasks [38], and works even zero-shot [39]. Later methods sharpen it: self-consistency votes over many sampled chains [40]. Inference-time reasoning then grew into a paradigm of its own with OpenAI o1 [41] and DeepSeek-R1 [15].

Reinforcement learning turns reasoning from a prompt into a trained skill. Reinforcement learning from human feedback (RLHF) can align language models with human preferences by optimising a learned reward model; InstructGPT performs this optimisation using PPO [42, 43]. When answers are checkable, a verifier can supply the reward. Verifiers were first trained for mathematical word problems [16], and DeepSeek-R1 [15] shows that RL with verifiable rewards alone can induce strong reasoning. Group Relative Policy Optimization (GRPO) [21] makes this practical by estimating advantages from groups of samples and dropping the value network; open implementations [44, 45] reproduce the SFT-then-GRPO pipeline.

These recipes need reasoning data, which is usually synthesised by a teacher model. STaR bootstraps rationales by keeping the ones that reach the right answer [18], and step-by-step distillation transfers a teacher's rationales to a smaller model [19]. HuatuoGPT-o1 [17] is the closest method to ours: a verifier first guides the model to synthesise correct CoT traces for SFT, then a second RL stage refines them. The recurring problem is the *quality* of the synthesised traces. Many reach the right answer by unfaithful reasoning, or fail; the usual fix is to keep verified traces and discard the rest, which throws away much of the pool. NTDH targets exactly this data-quality problem.

2.3. Positioning: Reasoning Meets Affect

These two lines have rarely met. On the affect side, reasoning has so far been used mostly at inference time, through prompting. THOR chains CoT steps to infer implicit sentiment [29]; recent work also adds decomposed reasoning for emotion-cause extraction [32]. None of these synthesises quality-controlled, RL-ready affective reasoning data, and none trains with a verifiable reward. On the reasoning side, the verifier-driven recipes above were built for mathematics and medicine. Their generic "rethink" steps are blind to the cues that govern affect: valence shifters, irony, idioms, the emotion/sentiment distinction, and the circumplex.

We sit at the intersection. To our knowledge, we are the first to bring the verifiable-reward reasoning recipe to comprehensive affective analysis through affect-specific synthesis, verification and refinement. Unlike HuatuoGPT-o1

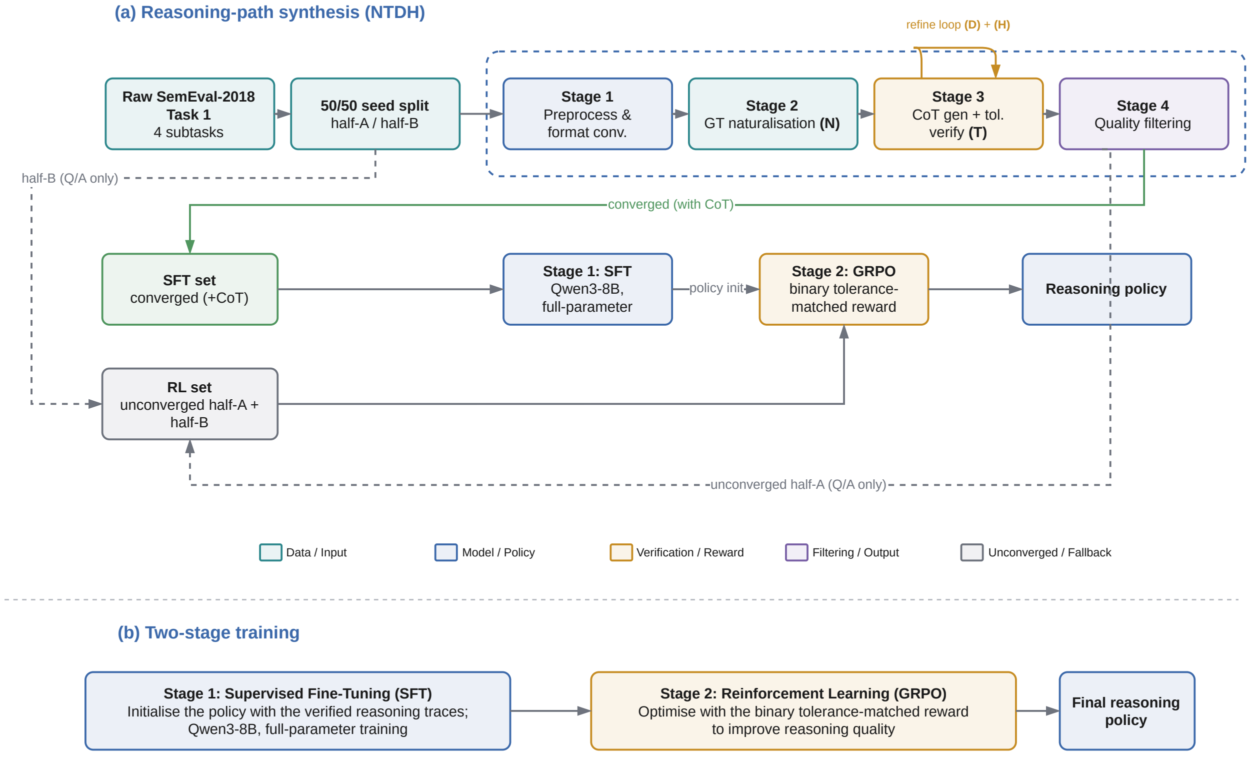

Figure 1: Overview of the proposed approach. NTDH turns raw SemEval-2018 data into verified, reasoning-augmented supervision through four synthesis stages. Naturalisation **(N)**, tolerance-aware verification **(T)**, domain-aware refinement **(D)** and directional hints **(H)** are realised in Stages 2–3. Converged traces supervise SFT; unconverged hard cases and half-B supply GRPO, yielding the final affective reasoning policy in one integrated pipeline.

[17], which discards unconverged cases, and EmoLLM [14], which is trained with SFT alone, we naturalise the gold label and route the hard cases into GRPO [21], so no training instance is wasted. §1 quantifies this claim.

3. Reasoning-Path Synthesis

We present **NTDH**, a quality-aware *reasoning-path synthesis* pipeline that converts raw SemEval-2018 labels into verified, reasoning-augmented training data. Inspired by HuatuoGPT-o1 [17], we adapt its SFT→GRPO framework to comprehensive affective analysis: SFT acquires the structured affective reasoning and prediction format from verified trajectories, and GRPO further improves the policy with task-aware rewards. We describe the training stage in §4 and defer all hyperparameters to §5.1.

NTDH builds on the EmoLLM Chain-of-thought (CoT) synthesis pipeline [14] and improves four of its stages, one per letter of the name:

- **(N) Naturalisation** rewrites each raw label as a scale-aware sentence, so the generator, the verifier, and the final supervised target all share one natural-language form (§3.1, Stage 2);
- **(T) Tolerance-aware verification** replaces the original binary LLM judge with a deterministic gate matched to each subtask's scoring tolerance (§3.3);
- **(D) Domain-aware refinement** grounds the four rethink strategies in the affective-science literature, so the refinement mirrors how affect is expressed and resolved (§3.4);
- **(H) Directional Hints** steer a stalled trajectory toward the gold answer without ever revealing it (§3.4).

Each surviving sample is tagged with a quality tier that drives the SFT/RL routing (§3.5). The end-to-end flow runs in four stages, shown in Figure 1. We first follow these stages in order (§3.1), then detail the task formulation, tolerance-aware gate (§3.3), domain-aware refinement and directional hints (§3.4), and quality-based routing (§3.5).

Table 1
Ground-truth naturalisation templates. Each subtask's raw label is converted into a sentence that explicitly states the task scale.

Task	Raw	Naturalised sentence
EI-reg	`0.562`	*Based on the text, the anger emotion has an intensity of 0.562 on a scale from 0 (lowest) to 1 (highest).*
V-reg	`0.532`	*The sentiment intensity of this text is 0.532 on a scale from −1 (most negative) to 1 (most positive).*
V-oc	2	*Intensity Class: 2: moderately positive mental state can be inferred.*
E-c	`7. anticipation, 9. optimism`	*The text's emotional tone is: 7. anticipation, 9. optimism.*

3.1. Data Processing Pipeline

We now walk through the four pipeline stages of Figure 1, from raw SemEval-2018 data to quality-filtered, reasoning-augmented training data; the improvements above are realised within Stages 2 and 3.

Stage 1: Preprocessing and format conversion. We convert the raw SemEval-2018 Task 1 data into a task-specific instruction that embeds the analysis objective and the expected output format, paired with the ground-truth label. The encodings differ by subtask: EI-reg names the target emotion and requests a score in $[0, 1]$; V-reg is rescaled to the symmetric scale $[-1, 1]$, so a stored value such as 0.532 already denotes a mildly positive valence; V-oc enumerates the seven ordinal classes ($-3 =$ very negative, …, $3 =$ very positive); and E-c uses a fixed index map over the eleven emotions (1. joy, …, 11. trust; 0. neutral when none applies; the ordering is fixed in the released code and is not the alphabetical listing of §3.2), applied identically in the gold target, the prompt, and the exact-set reward.

Stage 2: Ground-truth naturalisation. We convert each raw label into a fluent, scale-aware sentence with a per-subtask template (Table 1). The supervised <answer> target is set to this naturalised *ground truth*, not to the model's (possibly drifted) rewritten response; this is the single fix behind the answer-source correctness reported in §5.3. Because the conversion runs before CoT generation, the verifier compares model outputs against the same natural-language form as the final targets, removing one source of format mismatch.

Stage 3: CoT generation with verification. This is the core stage, summarised in Figure 2. A generator LLM produces an initial chain of *Inner Thinking* steps and a *Final Conclusion*, which the deterministic tolerance gate of §3.3 checks against the naturalised ground truth. On failure, the pipeline enters a two-level search: an outer loop of A fresh restarts (default A=3) and an inner loop of D refinement rounds (default D=3), each round applying one of the four domain-aware strategies of §3.4 and re-verifying. When the budget is exhausted, the fallback of §3.4 applies: hint refinement, allow failure, or label leakage. On convergence, the chain is reformatted into fluent, step-by-step reasoning, and a quality tier is assigned from the verification trajectory (§3.5).

Stage 4: Quality filtering and routing. The final stage routes samples as in §3.5: converged half-A samples (with CoT) form the SFT set; half-B and the unconverged half-A samples (question/answer only) form the RL set.

3.2. Task Formulation

We treat all four subtasks as a single conditional generation problem. Let x denote an input tweet paired with a task-specific instruction, and let the policy π_θ map x to a structured output containing an explicit reasoning trace and a final answer $\hat{y}$. The subtasks differ only in the form of $\hat{y}$ and the criterion by which it is scored:

- **EI-reg** (emotion-intensity regression): given a tweet and a target emotion $e \in \{\text{anger, fear, joy, sadness}\}$, predict an intensity $\hat{y} \in [0, 1]$.
- **V-reg** (valence regression): predict a real-valued sentiment $\hat{y} \in [-1, 1]$.
- **V-oc** (ordinal valence classification): predict one of seven ordinal classes $\hat{y} \in \{-3, \dots, 3\}$.
- **E-c** (multi-label emotion classification): predict the subset $\hat{y} \subseteq \mathcal{E}$ of present emotions, with $\mathcal{E} = \{$anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise, trust$\}$ (or neutral when none applies).

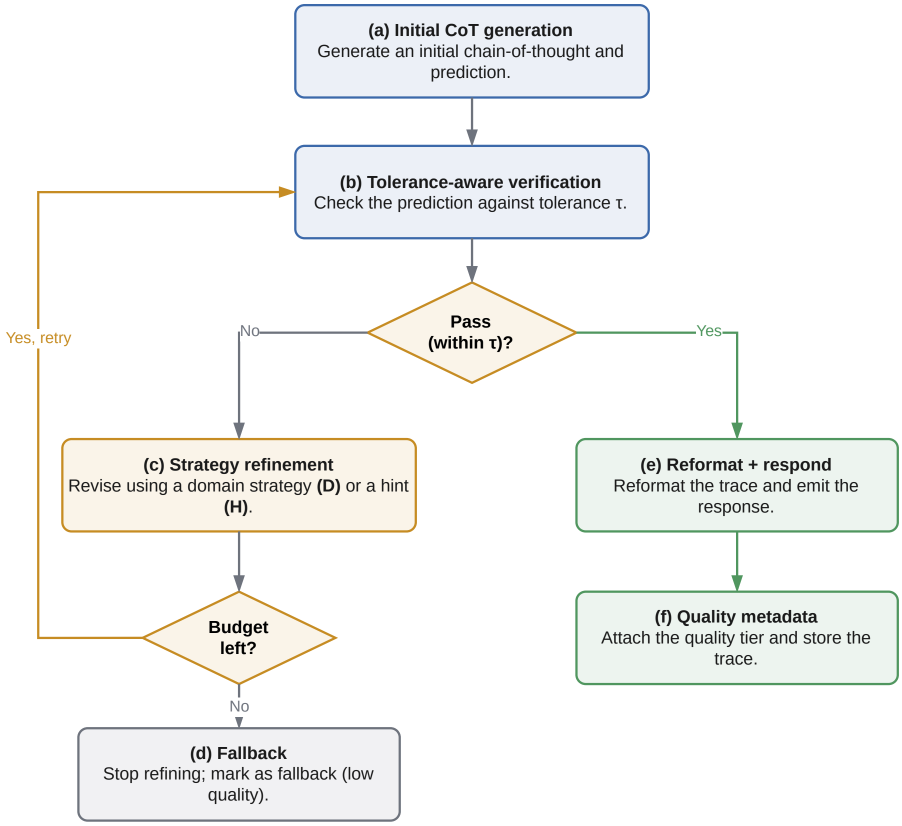

Figure 2: Inner loop of Stage 3 for a single sample. The refinement cycle (c → b) may iterate up to $D \times A$ times (where D is the search depth and A the number of restart attempts) before the fallback path (d) is taken.

A prediction is counted correct under a task-matched strict criterion τ: absolute error $|\hat{y} - y| \leq 0.05$ for the regression subtasks, exact class match for V-oc, and exact set match for E-c (Table 2). This τ defines the GRPO accuracy reward (§4) and the strict accuracy reported in our evaluation. The construction-time gate reuses it for the regression and ordinal subtasks, but is more permissive for E-c, where it accepts a trace at $F_1 \geq 0.7$ against the gold set (§3.3). Final reporting additionally follows the official SemEval metrics: Pearson correlation for EI-reg, V-reg, and V-oc, and Jaccard, micro-F1, and macro-F1 for E-c (§5.1). In Algorithm 1, component **(D)** samples one of four domain-aware refinement operations: *Backtrack* revisits an earlier interpretation, *Explore* tests an alternative cue path, *Verify* checks the draft against textual evidence, and *Correct* repairs a local label or intensity error; their affective grounding is detailed in §3.4. Here *Verify* is a reasoning-revision strategy within **(D)**, distinct from the deterministic accept/reject gate **(T)**. The per-sample synthesis procedure is illustrated on a concrete tweet in Figure 3. The fifth SemEval-2018 subtask, emotion-intensity ordinal classification (EI-oc), is a coarse re-binning of the EI-reg signal and is omitted as redundant.

In Algorithm 1, $\hat{y}$ is the generator's candidate conclusion and is used only to verify whether the associated reasoning chain satisfies the task-specific construction criterion. Once the chain converges, the naturalised gold answer y—rather than $\hat{y}$—is written to the supervised <answer> field.

3.3. Tolerance-Aware Verification

In the original EmoLLM construction pipeline [14], after the generator produces a CoT and final conclusion, a separate judge LLM receives that conclusion together with the reference answer and is prompted to return `True` (semantically consistent) or `False` (inconsistent). This judgement is a data-construction filter—it decides whether the trace is retained or refined—rather than the model's task prediction. Because the binary prompt does not define task scales or numerical tolerances, it can reject a valid regression near-match (e.g. 0.55 versus 0.56) or accept a value outside the evaluation tolerance. In the LLM-judge configuration we replaced, such false acceptance occurred for 48–63% of the out-of-tolerance regression responses presented to the judge, allowing incorrect targets into SFT.

Algorithm 1: NTDH reasoning-path synthesis (single sample)

Input: tweet+instruction x; gold answer y; tolerance τ; max restarts A, depth D
Output: reasoning trace and naturalised gold answer (SFT), or unconverged (RL)
$y \leftarrow$ NATURALISE(y) // (N) scale-aware gold sentence
(chain, $\hat{y}$) $\leftarrow$ GENERATECOT(x)
for $a \leftarrow 1$ **to** A **do**
  for $d \leftarrow 1$ **to** D **do**
    if $\hat{y}$ matches y within τ (T) **then**
      return REFORMAT(chain), y // gold y supervises the answer
    // (D) sample one domain-aware refinement operation
    $s \leftarrow$ SAMPLEDOMAINSTRATEGY({BACKTRACK, EXPLORE, VERIFY, CORRECT})
    $h \leftarrow$ DIRECTIONALHINT($\hat{y}$, y) // (H) label-free guidance
    (chain, $\hat{y}$) $\leftarrow$ REFINE(x, chain, s, h)
  reset chain
return unconverged // no CoT $\rightarrow$ RL set

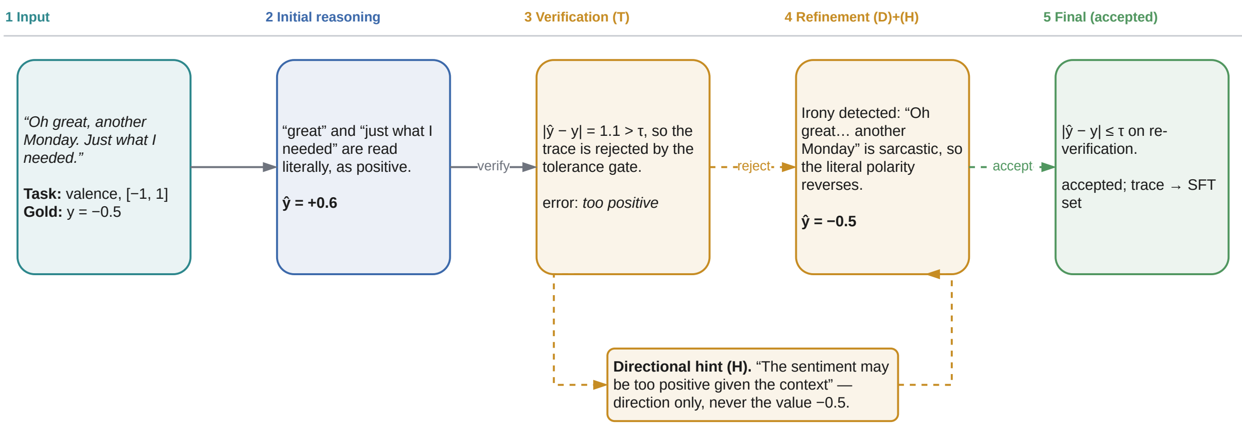

Figure 3: A worked example of one NTDH synthesis loop (illustrative). An ironic tweet is first read literally as positive. The tolerance-aware verifier **(T)** computes the task-specific error and rejects the trace; the directional hint **(H)** reports only that the prediction is too positive, without exposing the gold value; and *Backtracking* **(D)** then revisits the literal reading and detects the irony. The corrected trace passes verification and enters the SFT set.

NTDH separates format interpretation from the final decision. The judge LLM is retained only to parse heterogeneous natural-language conclusions and for logging; a deterministic tolerance-aware verifier **(T)** compares the parsed prediction with the gold answer under the task-specific construction criterion in Table 2. If the trace fails, the verifier passes only the signed error category—not the gold value—to the directional hint module **(H)**. Thus, in Figure 3, **(T)** computes $|0.6 - (-0.5)| = 1.1 > 0.05$ and rejects the trace, whereas **(H)** communicates only "too positive". The domain-aware strategy **(D)** then determines how to revise the reasoning, here by backtracking to inspect irony. For E-c only, the construction gate accepts $F_1 \geq 0.7$ so a mostly correct reasoning trace can supervise SFT, whereas the GRPO accuracy reward requires an exact label-set match. Because Naturalisation pairs an accepted trace with the complete gold label set, a trace that passes this permissive gate may not explicitly justify every label in its supervised answer. This reasoning–answer mismatch is a limitation of the present construction and is discussed in §6.

3.4. Domain-Aware Refinement and Directional Hints

When the initial CoT fails verification, the pipeline enters an iterative refinement loop. The original framework either randomly selects from four generic search strategies or falls back to *label leakage*, where the ground-truth

Table 2
Task-specific criteria used by the deterministic construction verifier and the GRPO accuracy reward. The E-c construction gate is intentionally more permissive. Final reporting additionally follows the official SemEval metrics listed in §5.1.

Task	Scale	Construction verifier (T)	GRPO strict criterion
EI-reg	[0, 1]	Absolute error ≤ 0.05	Same
V-reg	[−1, 1]	Absolute error ≤ 0.05	Same
E-c	11 labels	Label-set $F_1 \geq 0.7$	Exact label-set match
V-oc	7 classes	Exact class match	Same

answer is injected into the prompt. Label leakage guarantees convergence at the cost of superficially correct reasoning that contaminates the training data.

Domain-aware search strategies. The four strategies—*Backtracking*, *Exploring New Paths*, *Verification*, and *Correction*—are each augmented with sentiment- and emotion-specific guidance grounded in a finding from affective science, so the refinement mirrors how affect is expressed and resolved in text rather than following a generic "rethink". *Backtracking* fires when a deeper cue overturns the surface reading: emotion and sentiment differ [2], valence shifters (negation, intensifiers, diminishers) can flip or rescale a valenced word [3], and irony reverses the literal affect [5]. *Exploring New Paths* switches from lexical to contextual cues—idioms and metaphors are non-compositional [6], emotions co-occur in regular (Plutchik) patterns [9], and Russell's circumplex [7] spreads affect over arousal × valence. *Verification* checks the draft label directly against evidence in the tweet, including emotion words, shifters, emoji, and punctuation. *Correction* fixes fine-grained, often ordinal errors—degree adverbs give an intensity order (*slightly* < *very* < *extremely*) [3], the anger–disgust co-occurrence [8] licenses targeted E-c fixes, and the contextual-versus-prior polarity distinction [4] guides the V-oc thresholds. The strategies are applied *one at a time across successive refinement rounds*: each round randomly selects one, appends its reasoning, and re-verifies (the inner loop of Stage 3, §3.1).

Hint mode. Instead of revealing the label, the refinement prompt receives only a *directional hint* derived from the verifier. We design task-specific hint logic rather than a uniform error-based scheme:

- **Regression tasks** (EI-reg, V-reg): the hint indicates whether the prediction is *too high*, *too low*, *slightly high*, *slightly low*, or *close*, calibrated by error magnitude, with the band cut-offs given in the released code; on the signed valence scale these are rendered as *too positive/too negative*.
- **Multi-label classification** (E-c): the hint conveys *over-identified* (too many emotions), *incomplete* (missed emotions, with co-occurrence reminders such as anger–disgust), or *partially correct*.
- **Ordinal classification** (V-oc): instead of a generic "incorrect" signal, the hint specifies sentiment polarity direction with three granularity levels: *slightly too positive/negative* (one-level deviation), *too positive/negative* (two- to three-level), and *far too positive/negative* (four or more levels).

The model is steered toward the correct answer without ever seeing the label, so the resulting reasoning chain stays authentic.

Failure handling. After exhausting the search budget, three mutually exclusive strategies are available: (1) *hint refinement*, which provides a directional hint and re-attempts verification without forcing success (recommended); (2) *allow failure*, which marks the sample as unconverged and excludes it from the SFT set (it is still usable for RL, §3.5); or (3) *label leakage* (legacy), which injects the ground-truth label; such samples are flagged and assigned the lowest quality tier.

Enhanced search budget. We raise the maximum number of fresh restarts from 1 to 3 and the refinement depth per attempt from 2 to 3, giving the search more room to converge without resorting to label leakage.

3.5. Quality Tiering and Data Routing

Each generated sample is assigned a quality tier based on its verification trajectory (Table 3). The corresponding quality weight can be used for weighted loss during supervised fine-tuning or for data selection.

Here the quality tier mainly serves as a converged/unconverged filter. Inspired by the two-stage design of HuatuoGPT-o1 [17], we route verified reasoning traces to SFT and answer-labelled cases to GRPO. We split each

Table 3
Quality tier definitions. *Iterations* denotes the number of verification rounds before the first success.

Tier	Wt.	Iter.	Description
Gold	1.0	1	First-attempt success
Silver	0.8	2–3	Success after limited refinement
Bronze	0.5	≥4	Success after extensive refinement
Low	0.0	—	Success via label leakage
Unconv.	0.0	—	No correct answer reached

subtask's training pool 50/50 by a fixed seed (42) into *half-A* and *half-B*. Half-A is sent through the CoT pipeline above; its *converged* samples (gold/silver/bronze, all carrying a non-empty reasoning chain and an answer) form the **SFT** set. The **RL** set is the union of (i) all of half-B, which is never sent through CoT generation and so carries only (question, answer) pairs, and (ii) the *unconverged* hard samples from half-A. GRPO can still extract a reward signal from a (question, answer) pair without a CoT trace. Under this routing every training instance is used exactly once (§5.1). Weighted loss using the gold/silver/bronze sub-weights is left as future work; in the present setup all retained samples enter training with uniform weight.

4. Two-Stage Training

We train Qwen3-8B in two stages and defer all hyperparameters to §5.1.

Stage 1: Supervised fine-tuning. We fine-tune Qwen3-8B [20] with full-parameter SFT on the converged, reasoning-augmented SFT set, teaching the policy to emit a structured <think> trace followed by an <answer>. This stage fixes the reasoning format and yields a usable initial policy, whose checkpoint initialises the RL stage.

Stage 2: Reinforcement learning with GRPO. Starting from the SFT policy, we apply GRPO [21] on the RL set with a binary, tolerance-matched reward (§5.1): a sampled completion scores 1 when its parsed answer meets the task tolerance τ of §3.2 and 0 otherwise. GRPO estimates advantages from groups of sampled completions, so it needs no separate value network. It learns directly from the strict criterion of Table 2, which also gates construction on the regression and ordinal subtasks, closing the loop between synthesis and optimisation. The unconverged hard cases (a gold answer but no CoT trace) supply additional reward signal at this stage rather than being discarded.

5. Experiments and Results

5.1. Experimental Setup

We run the full pipeline with all four NTDH improvements enabled.

Data. We build on SemEval-2018 Task 1 [10], restricted to the four English subtasks (E-c, EI-reg, V-reg, V-oc). Each subtask's training pool is halved by a fixed seed (§3.5): half-A is sent through CoT synthesis and its converged samples become the SFT set, while half-B together with the unconverged half-A hard cases form the RL set. Every training instance is therefore used exactly once (100% utilisation; 16,302 instances total: 5,388 SFT and 10,914 RL), and Table 4 gives the per-subtask split sizes.

Realised quality-tier distribution. Table 5 reports the quality tiers (§3.5) assigned to the 8,150 half-A CoT generations, computed from each sample's verification trajectory. *No* sample required label leakage (the *low* tier is empty): directional hints (§3.4) replaced the gold-answer injection that the original framework fell back to.

Baselines. We situate our model against the prior systems collected in the EmoLLM benchmark [14], in four groups: (i) the SemEval-2018 competition systems SeerNet [26] and NTUA-SLP [27]; (ii) fine-tuned pre-trained language models (BERT, RoBERTa, SentiBERT); (iii) zero- and few-shot LLM baselines (Falcon, Vicuna, LLaMA2, ChatGPT, GPT-4); and (iv) the instruction-tuned EmoLLM family (EmoBART/T5/OPT/BLOOM and EmoLLaMA), of which

Table 4
Dataset statistics per subtask. Half-A is sent through CoT generation; its converged samples form the SFT set, and its hard (2,746 unconverged and 16 empty- or invalid-chain) samples join half-B in the RL set. All reported results use the official test split.

Subtask	half-A (CoT)	SFT (conv.)	→RL (hard)	half-B (RL src)	RL (total)	official (test)
EI-reg	3,551	2,585	966	3,551	4,517	4,068
E-c	3,419	1,999	1,420	3,419	4,839	3,259
V-reg	590	315	275	591	866	937
V-oc	590	489	101	591	692	937
Total	8,150	5,388	2,762	8,152	10,914	9,201

Table 5
Realised quality-tier distribution of the 8,150 half-A CoT generations, assigned by verification trajectory (§3.5). The gold/silver/bronze tiers contain 5,404 converged trajectories; 16 empty- or invalid-chain cases are routed to RL, leaving 5,388 SFT trajectories. The *low* (label-leakage) tier is empty under the hint-refinement fallback.

Task	Gold	Silver	Bronze	Unconv.	Total
EI-reg	371	1,477	744	959	3,551
E-c	914	828	260	1,417	3,419
V-reg	45	186	85	274	590
V-oc	169	262	63	96	590
All	1,499	2,753	1,152	2,746	8,150

EmoLLaMA-chat-13B is the strongest on EI-reg and V-oc and serves as our primary reference. All baseline numbers are reproduced verbatim from [14] and are scored on the official test split. Table 7 reports their results and our official-test results in the same task-specific format.

Metrics. We report the SemEval-2018 primary metrics, so that our results are directly comparable with prior work:

- **EI-reg, V-reg** (regression) and **V-oc** (7-class ordinal): Pearson correlation r against the gold values, with EI-reg macro-averaged over the four target emotions.
- **E-c** (11-label multi-label): Jaccard similarity (the SemEval "acc"), micro-F1, and macro-F1.

The strict, tolerance-based accuracy of Table 2 is used for the GRPO reward but is not reported as a headline metric, since no prior system reports it. Predictions are parsed from the structured <think>/<answer> output produced by the policy; there are no parsing failures for either checkpoint reported on the official test.

Models and training. We fine-tune **Qwen3-8B** [20] with full-parameter SFT, then run GRPO [21] from the SFT checkpoint with the open-r1 / TRL stack [44, 45]; Table 6 lists the hyperparameters. We report two checkpoints on the official test: the SFT checkpoint used to initialise GRPO (*SFT init*, step 280) and the last saved GRPO checkpoint (*RL-final*). Throughout, "step" denotes the trainer's logged global step. Both were fixed in advance; no checkpoint selection was performed on the official test split.

Reward function. The GRPO reward combines a binary accuracy term with a small format bonus, $1.0 \cdot r_{\text{acc}} + 0.1 \cdot r_{\text{fmt}}$. The accuracy reward r_{acc} is *binary*: it is 1 when the parsed prediction satisfies the task's *GRPO strict criterion* in Table 2—note that this requires an exact label-set match on E-c, unlike the permissive construction gate—and 0 otherwise; a missing <answer> block scores 0. The format reward r_{fmt} is 1 iff the output matches the <think>...</think>\n\n<answer>...</answer> template. The consequences of this binary objective are examined through the official-test error analysis in §6.

Table 6
Training hyperparameters for the two stages, both fine-tuning Qwen3-8B. "—" marks settings that do not apply.

	SFT	**GRPO**
Framework	LLaMA-Factory	open-r1 / TRL
Learning rate	$5{\times}10^{-6}$	$1{\times}10^{-6}$
LR schedule	cosine, 0.1 warmup	cosine, 0.1 min-LR
Epochs	3	2
Per-device batch	1 (×2 GPUs)	2
Gradient accumulation	8	4
Max length	8192	4096 / 8192 (prompt / completion)
Precision / runtime	bf16, ZeRO-3	vLLM colocated generation
Validation split	10% held out	—
Group size G	—	8
KL coefficient β	—	0.01
Sampling temperature	—	1.0

5.2. Main Results and Comparison with Prior Work

Official-test comparison. The official NTDH scores and prior-system results are compared directly in Table 7. NTDH achieves its strongest result on EI-reg. RL-final obtains a macro-averaged Pearson correlation of 0.862, outperforming SeerNet by 0.063 and EmoLLaMA-chat-13B by 0.031. It also achieves the highest correlation on all four target emotions. Performance is more mixed on the other subtasks. RL-final reaches 0.840 on V-reg and 0.831 on V-oc, trailing the strongest baselines by 0.047 and 0.029, respectively, while its E-c Jaccard of 0.579 is the second-best score in the table.

Comparison within the EmoLLM family. NTDH remains competitive with the EmoLLM family while using approximately 14× fewer total training records than EmoLLM's AAID corpus. RL-final exceeds all seven EmoLLM variants on EI-reg, while its V-reg and V-oc results remain below the strongest members of that family. On E-c it achieves a Jaccard score of 0.579, exceeding the strongest EmoLLM result, EmoT5 at 0.559, by 0.020, while remaining 0.030 below SeerNet. This pattern indicates that the compact NTDH corpus is effective for emotion-intensity regression and structured multi-label prediction, although macro-F1 shows that rare-label recall remains a limitation.

Two-stage training outcome. Relative to the SFT initialisation checkpoint on the same official split, RL-final improves EI-reg Pearson correlation from 0.800 to 0.862, V-reg from 0.785 to 0.840, and V-oc from 0.785 to 0.831. On E-c, Jaccard increases from 0.557 to 0.579 and micro-F1 from 0.667 to 0.677, whereas macro-F1 decreases from 0.555 to 0.543. The EI-reg improvement occurs across all four target emotions: +0.076 for anger, +0.059 for fear, +0.034 for joy, and +0.080 for sadness.

5.3. Ablation of NTDH Components

We isolate each NTDH component and measure its effect on the synthesis output, scored against the true tolerance τ of §3.2. These figures are data-side: they characterise the quality of the synthesised data independently of downstream model evaluation. For N, T and H, the effect is an *offline counterfactual* read off the full synthesis run; for D, which is on by default, we run a matched 320-instance re-synthesis with generic rethink prompts. Table 8 summarises the results.

Table 7
Comparison with prior work on SemEval-2018 Task 1 (English). EI-reg / V-reg / V-oc use Pearson correlation r; the EI-reg "ave" column is the macro-average over the four emotions. E-c is reported with multi-label accuracy ("acc", i.e. Jaccard), micro-F1 and macro-F1. All systems, including the two NTDH checkpoints, are evaluated on the official test split. The best score in each column is in **bold**, and the second-best score is underlined; ties receive the same marking. EI-oc is excluded from our pipeline by design, so its columns are omitted.

	EI-reg (Pearson)					**V-reg**	**V-oc**	**E-c**		
System	ave	anger	fear	joy	sadness	(Pearson)	(Pearson)	acc	mi-F1	ma-F1
SemEval-2018 competition systems										
SemEval baseline	0.520	—	—	—	—	0.585	0.509	—	—	—
NTUA-SLP	0.757	—	—	—	—	0.839	0.765	0.579	—	—
SeerNet	0.799	0.827	0.779	0.792	0.798	0.873	0.856	**0.609**	**0.724**	**0.592**
Pre-trained language models										
BERT-base	0.785	0.800	0.781	0.783	0.742	0.840	0.805	0.567	0.718	0.568
RoBERTa-base	0.717	0.670	0.736	0.769	0.694	0.845	0.772	0.563	0.721	0.536
SentiBERT	0.722	0.724	0.740	0.731	0.691	0.835	0.763	0.535	0.700	0.522
Zero- / few-shot LLM baselines										
Falcon	0.114	0.147	0.082	0.095	0.131	0.135	0.189	0.190	0.318	0.253
Vicuna	0.281	0.307	0.257	0.260	0.299	0.298	0.579	0.220	0.359	0.253
LLaMA2-7B-chat	0.194	0.176	0.257	0.097	0.247	0.094	0.497	0.257	0.414	0.286
LLaMA2-13B-chat	0.488	0.524	0.506	0.398	0.526	0.312	0.568	0.274	0.424	0.302
ChatGPT	0.599	0.637	0.573	0.569	0.618	0.637	0.748	0.382	0.546	0.429
ChatGPT-FS	0.550	0.572	0.482	0.587	0.560	0.739	0.791	0.413	0.563	0.466
GPT-4	0.656	0.699	0.575	0.686	0.667	0.811	0.788	0.444	0.572	0.497
GPT-4-FS	0.679	0.704	0.654	0.679	0.678	0.825	0.793	0.460	0.582	0.515
EmoLLM family (instruction tuning on AAID)										
EmoBART	0.795	0.798	0.803	0.795	0.782	0.851	0.835	0.528	0.686	0.548
EmoT5	0.783	0.785	0.797	0.798	0.751	0.852	0.836	0.559	0.712	0.568
EmoOPT	0.825	0.827	0.830	0.837	0.805	**0.887**	0.843	0.532	0.680	0.550
EmoBLOOM	0.791	0.802	0.797	0.790	0.776	0.857	0.822	0.528	0.683	0.552
EmoLLaMA-7B	0.822	0.819	0.821	0.837	0.809	0.879	0.843	0.545	0.695	0.563
EmoLLaMA-chat-7B	0.824	0.825	0.830	0.832	0.810	0.876	0.827	0.534	0.693	0.540
EmoLLaMA-chat-13B	0.831	0.827	0.835	0.843	0.817	0.886	**0.860**	0.537	0.696	0.545
Ours (NTDH CoT → SFT → GRPO on Qwen3-8B)										
SFT init (ckpt-280)	0.800	0.753	0.799	0.845	0.802	0.785	0.785	0.557	0.667	0.555
RL-final	**0.862**	**0.829**	**0.858**	**0.879**	**0.882**	0.840	0.831	0.579	0.677	0.543

Table 8: Ablation of the four NTDH components, each removed in isolation. Metrics are data-side, measured on the CoT-synthesis output against the true tolerance. For N, 18.4% is the proportion of initial model conclusions within gold tolerance over all 8,150 half-A samples; 100% denotes correctness of the gold-derived answer field for retained SFT traces. T and H are computed on the full run and D on a matched 320-instance sample.

Component removed	**Data-quality metric**	**NTDH**	**w/o**
Naturalisation (N)	answer-source gold consistency	100%	18.4%
Tolerance gate (T)	out-of-tol. accepts (EI-reg / V-reg)	0%	62.7%/47.5%
Domain strat. (D)	convergence yield (overall / E-c)	66%/56%	65%/46%
Hints (H)	label leakage in the fallback	0%	gold answer injected

Naturalisation (N). Without naturalisation, the candidate <answer> is the model's own initial conclusion, which lands within gold tolerance for only 18.4% of the 8,150 half-A samples, and far less often on the regression subtasks. With naturalisation, the answer source is instead the naturalised gold label, so the <answer> field of every retained SFT sample is correct by construction. Accordingly, 100% here denotes answer-target correctness. The synthesis stage retains 5,388 samples with verified, non-empty traces for SFT and routes the other 2,762 half-A samples to RL.

Tolerance gate (T). Replacing the deterministic gate with the original LLM judge silently admits out-of-tolerance regression answers most of the time (Table 8), corrupting the supervised targets; the ordinal and multi-label subtasks are less affected. The deterministic gate removes this leakage entirely.

Domain-aware strategies (D). Domain strategies barely move the overall convergence yield, but the effect is task-dependent: they help the fine-grained, multi-label E-c subtask by about 10 points (Table 8), where Plutchik co-occurrence and the circumplex apply, while leaving the coarser regression subtasks unchanged. Their role is to keep the refinement grounded in affective-science theory rather than to raise yield, consistent with reasoning mattering most at the micro level. These are single-run estimates, so the small overall gap may reflect run-to-run variation; the larger E-c difference is promising but requires repeated runs before it can be treated as robust.

Directional hints (H). When a trace exhausts the search budget, the original framework fell back to injecting the gold answer, so any trace rescued that way carried the target in its own reasoning. Directional hints replace that fallback with a label-free signal, and the effect is visible in the realised tier distribution: the *low* (label-leakage) tier of Table 5 is empty, so *no* retained trace owes its supervision to a leaked answer (Table 8). Traces that the hints do not rescue are routed to RL rather than repaired, so the mechanism trades a lower yield for supervision that is honestly derived throughout.

6. Discussion

Target quality under sparse verifiable rewards. The final two-stage policy scores above the SFT initialisation checkpoint on five of six official-test metrics while using 16,302 total SFT and RL records, approximately 14× fewer than the ~234K records in EmoLLM's AAID corpus. The SFT stage itself uses only 5,388 reasoning trajectories. This result is consistent with the importance of target quality when reward is sparse. With a $\{0, 1\}$ reward and group-relative advantages, a prompt group yields no GRPO gradient when all its completions score alike. A controlled estimate obtained with a proxy generator rather than the fine-tuned policy ($n = 70$ prompts, $k = 8$ samples per prompt) places the share of such zero-variance groups at 77.1% for E-c and 75.7% for V-reg; RL thus learns only from the minority of groups that straddle the decision boundary, while wrong SFT targets anchor the policy badly before RL even begins. Naturalisation replaces the model-derived answer source (only 18.4% of initial conclusions meet the gold tolerance) with gold-consistent targets for every retained SFT sample. Together with the tolerance-aware gate, it is therefore a main lever; adding data mostly adds more zero-variance groups. The one official-test metric that declines, E-c macro-F1 ($0.555 \rightarrow 0.543$), motivates the qualitative error analysis below.

Error analysis on the official test. The representative cases in Table 9 expose three recurring failure modes: reasoning–answer inconsistency on E-c, collapse of mixed-valence clauses into neutrality on V-oc, and over-correction on explicit sarcasm markers. The remaining errors therefore arise from rare-label omission, fine-grained intensity calibration, and conflicting contextual cues rather than from output formatting.

Limitation of permissive E-c verification. The $F_1 \geq 0.7$ E-c construction threshold improves trace coverage but does not guarantee that an accepted trace supports every label in the complete gold answer supplied through Naturalisation. Consequently, some E-c SFT records may contain a locally plausible reasoning trace paired with a broader gold label set. Our present verifier checks the candidate conclusion against the gold set but does not separately test entailment between each reasoning step and each output label. The results should therefore not be interpreted as evidence that every generated rationale is fully faithful. A label-wise reasoning–answer consistency gate is an important next step.

Implications of the cases. The E-c example shows that a plausible trace is insufficient unless the final answer is checked against the trace itself; future verification should therefore add a reasoning–answer consistency test before

Table 9
Representative errors of the GRPO policy on the official test. Gold and predicted labels are shown in the benchmark's output space; excerpts are shortened only for presentation.

Task	Tweet excerpt	Gold → prediction	Observed failure
E-c	"Indian army is on its way to dispatch all terrorists to hell"	{anger, anticipation, optimism, trust} → {anger, disgust}	The trace mentions anticipation and optimism, but the final answer omits them and substitutes disgust, exposing a reasoning–answer consistency error.
V-oc	"YAY, that's awesome ... it makes me sad you're leaving"	$-2 \rightarrow 0$	Opposing positive and negative clauses are collapsed into neutrality instead of resolving which cue controls the ordinal judgement.
V-reg	"The fun never stops. [upside-down-face] #sarcasm"	$0.392 \rightarrow -0.266$	The explicit sarcasm marker triggers a polarity reversal stronger than the benchmark annotation, illustrating over-correction by a salient domain cue.

applying the gold-based gate. The mixed-valence V-oc case motivates an explicit cue-weighting step, whereas the sarcasm case cautions against treating a domain hint as a deterministic label flip. Together, the cases favour confidence-aware, label-specific rewards over uniformly stronger refinement.

Two lessons for verifiable-reward pipelines. First, the binary reward decouples strict accuracy from the continuous metrics by construction: once a prediction enters the ± 0.05 band, no additional credit distinguishes a close estimate from an exact one. A shaped reward (soft F1 for E-c and distance-aware credit for regression and ordinal tasks) is therefore a natural next step. Second, multi-stage SFT→RL routing changes dataset membership after the initial split, so split disjointness has to be re-verified after every routing decision rather than once at the start.

7. Conclusion

We asked whether a small, carefully synthesised reasoning corpus plus RL can make an 8B model competitive on fine-grained affective analysis without the hundreds of thousands of supervised instances prior recipes use. NTDH answers the data-quality half of the question: ground-truth naturalisation and a tolerance-aware gate provide gold-consistent answer targets for all 5,388 retained SFT traces, while routing the other 2,762 half-A cases to RL uses all 16,302 instances. A component ablation (§5.3) isolates the contribution of each stage. On the official 9,201-instance test set, RL-final obtains Pearson correlations of 0.862, 0.840, and 0.831 on EI-reg, V-reg, and V-oc, respectively, together with Jaccard, micro-F1, and macro-F1 scores of 0.579, 0.677, and 0.543 on E-c. It scores above the SFT initialisation checkpoint on five of six metrics and achieves the strongest EI-reg result among the compared systems, demonstrating competitive performance across four heterogeneous subtasks (Table 7). Two refinements follow from our analysis: a shaped GRPO reward that grades how close a prediction lies to the scoring tolerance, and a quality-weighted SFT loss that uses the gold/silver/bronze tiers already produced during synthesis.

Beyond SemEval-2018, NTDH is a general recipe for building verifiable-reward reasoning data. Its two core ideas (deriving the verification gate and the reward from one task tolerance, and steering failed traces with label-free directional hints) apply to any task with a checkable answer, such as graded relevance, ordinal stance, or multi-label tagging. NTDH therefore applies to verifiable-reward training on information-retrieval and text-mining tasks, not only on affect.

Code and Data Availability

The NTDH data pipeline, the (frozen) SFT/GRPO training configurations, the evaluation harness, the released prediction and metric files, and the official-test inference script are organised in a single self-contained repository, to be released publicly upon publication. The two training files (SFT and RL) rebuild bit-identically from the raw SemEval-2018 inputs at the fixed seed reported above.

Ethical Considerations

SemEval-2018 Task 1 is a public benchmark of English tweets, used here under its original terms; the excerpts in Table 9 are included solely for aggregate error analysis and are minimally reproduced. Affective analysis of social-media text carries dual-use risk: the same models that can support, e.g., mental-health screening can also enable intrusive surveillance of individuals' emotional states. We therefore frame this work as a study of *reasoning and data quality* on a fixed academic benchmark, and caution that any deployment on real users requires consent, privacy safeguards, and domain-appropriate validation well beyond the benchmark evidence presented here.

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